

1. 1. Let G be a plane graph with V vertices, E edges, F faces and C components. Prove that

$$V - E + F - C = 1$$

Induction on C :

Base case: If there is only 1 component, then we have, by Euler's formula, that

$$V - E + F - C = V - E + F - 1 = 1$$

Induction: Assume that $V - E + F - C = 1$ holds for all planar graph components with fewer than k components. Let there be k planar graph components. An edge is then added between two of the components to connect them this decreases the amount of components by 1. This edge will add 1 to E and subtract 1 from C , but will not change V or F . Thus, we have that

$$V - (E + 1) + F - (C - 1) = 1$$

which simplifies to

$$V - E + F - C = 1$$

Thus, by induction, we have that $V - E + F - C = 1$ for all planar graphs.